

JOYDEP DHAR

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RESEARCH INTERESTS

Computer Vision · Deep Learning · Anomaly Detection · Real-Time Surveillance Systems · Convolutional Neural Networks

EDUCATION

American International University-Bangladesh (AIUB)

2020 – 2024
CGPA: 3.80 / 4.00

B.Sc. in Computer Science & Engineering (CSE)

- Graduated with consistent academic excellence; Dean's List honoree for four consecutive semesters (Fall 2021–22 to Spring 2022–23).
- Completed a research thesis on deep learning-based real-time criminal activity detection.

Shahid Syed Nazrul Islam College, Mymensingh

2019
GPA: 4.92 / 5.00

Higher Secondary Certificate (HSC)

RESEARCH EXPERIENCE

Undergraduate Thesis Research

2023 – 2024

American International University-Bangladesh(AIUB), Supervisor: Md. Faruk Abdullah Al Sohan, Former Assistant Professor, Dept. of Computer Science, AIUB

Criminal Activity Detection in Real-Time Footage: Leveraging Machine Learning for Accurate Criminal Activity Recognition

- Proposed and implemented a deep learning pipeline to detect and classify anomalous activities in real-time CCTV surveillance footage across **14 crime categories** (Abuse, Arrest, Arson, Assault, Burglary, Explosion, Fighting, Road Accidents, Robbery, Shooting, Shoplifting, Stealing, Vandalism, Normal).
- Trained and benchmarked **ResNet-50** against VGG-19 on the **UCF Crime Dataset** (777,653 frames; 85.7% train / 14.3% test split), achieving **91.56% test accuracy** and 76.97% validation accuracy — significantly outperforming VGG-19 (58%).
- Applied image pre-processing: brightness, contrast, and sharpness normalization with geometric augmentation (rotation $\pm 20^\circ$, zoom, flip, shear) and pixel scaling to $[0, 1]$.
- Evaluated using Precision, Recall, F1-Score, and AUC-ROC curves; ResNet-50 exceeded 90% per-class accuracy on almost all 14 crime classes.
- Validated in real-world scenarios by running the trained model on live CCTV test footage, correctly classifying arrest, robbery, shoplifting, explosion, abuse, and normal events in real time.
- Established that ResNet-50's residual skip connections effectively resolve the vanishing gradient problem, outperforming VGG-19 in both accuracy and training efficiency on identical hardware.
- Reviewed 19 prior crime-detection systems, identifying key limitations and positioning the proposed approach against current state-of-the-art benchmarks.

TECHNICAL SKILLS

Deep Learning & CV: ResNet-50, VGG-19, CNN, TensorFlow, Keras, Transfer Learning, Image Augmentation

Programming: Python, JavaScript, C#, C++, Bash

Frameworks & Libraries: Django, Django REST Framework, ASP.NET, React, Tailwind CSS

Data & Visualization: Power BI, Pandas, NumPy, Matplotlib

Databases: MySQL, PostgreSQL, SQLite

Tools & Platforms: Git, GitHub, Odoo ERP, Google Colab, Vercel, Render

Operating Systems: Ubuntu Linux, Windows

PROFESSIONAL EXPERIENCE

Python Developer — Zylo, Dhaka

July 2025 – Present

- Developing and customizing Odoo ERP modules to automate workflows across manufacturing, sales, and inventory.
- Building backend logic, integrating REST APIs, and optimizing performance in production ERP environments.
- Managing Git-based version control, CI/CD deployment pipelines, and production debugging.

VOLUNTARY EXPERIENCE

IT Support Volunteer — Nirvoy Foundation

February 2025 – June 2025

Non-Profit Organization, Bangladesh

- Provided IT infrastructure support and technical assistance for the organization's digital operations.
- Designed and executed digital marketing campaigns and graphic design materials to strengthen online outreach.
- Managed social media channels and produced visual content, increasing community engagement for a public welfare cause.

SELECTED PROJECTS

E-Commerce Platform

Python, Django, React

- Full-stack platform with JWT authentication, product management, cart, and order/payment workflows; deployed via Vercel and Render with CI/CD.

Learning Management System (LMS)

Python, Django REST Framework, React

- Role-based LMS for students, teachers, and admins with course enrollment, real-time progress tracking, and completion analytics.

Social Media Application

Python, Django

- Social platform with authentication, post creation, dynamic feeds, likes, and comments using Django ORM and Bootstrap.

CERTIFICATIONS & AWARDS

- **Dean's List Honor Award** — AIUB (Fall 2021–22 to Spring 2022–23, four consecutive semesters)
- **CISCO IT Essentials Certification** — Cisco Networking Academy
- **Full Stack Web Development** (Python–Django–React) — Ostad Limited
- **Crash Course on Python** — Google (Coursera)
- **Introduction to Data Science** — Simplilearn Skillup

REFERENCES

Md. Faruk Abdullah Al Sohan

Thesis Supervisor

Former Assistant Professor Dept. of CSE — American International University-Bangladesh

Graduate Research Assistant

UM Power Energy Dedicated Advanced Centre (UMPEDAC)

University of Malaya, Malaysia.

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